

**Sri Lanka Institute of Information
Technology**



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Web-Based Supermarket Ordering System

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Project Overview

Quickmart is a rapidly growing supermarket chain in Sri Lanka that continues to expand its operations by embracing the latest technological advancements.

A web-based supermarket ordering system is required to streamline the online shopping experience by allowing customers to browse products, place orders, make secure payments and receive home deliveries.

This system mainly focuses on order processing, inventory tracking, delivery coordination and secure payment handling to optimize operational efficiency.

Requirement Gathering Process

We employed a clear and organized requirement gathering procedure to develop a practical and effective web-based supermarket ordering system.

Defined the Scope of the System

We outlined the system's core functional areas, including **user** management, product catalog, order handling, payment integration, delivery coordination, and reporting. This helped us stay focused on the primary goals throughout the planning stage.

Using Microsoft CoPilot to Aid the Process

We used Microsoft CoPilot as a virtual assistant to generate stakeholder-specific questions, identify business needs, and explore potential features for each user type.

We also used it to roleplay different users to get a better understanding of the stakeholders.

Identified Functional & Non-Functional Requirements

- **Functional Requirements:** e.g., placing an order, updating product stock, assigning deliveries, processing payments.
- **Non-Functional Requirements:** e.g., system performance, ease of use, security, and scalability.

Structured Requirements by User Roles

Team members identified the responsibilities assigned to each team member for specific sections like inventory, payments, customer interaction and delivery

Reviewed and Finalized the Requirements

Conducting a team-wide review to refine all documented requirements.

We are Doing a Web Based Supermarket ordering System

 Project Discussion Bot - Year2 ⓘ

Great! Thanks for sharing that. Before we begin, I need a few more details to understand your project better.

Can you please tell me:

1. What is the title of your project?
2. Who will use this system? (e.g., customers, staff, admin)
3. What does the system do? (Just a few lines is enough)

Once I know this, I will create some client personas for you to talk to. Let's make this fun and useful! 😊

     

1.QuickMart Ordering System
2.Customer, Staff, Admin, Delivery
3. QuickMart is a local supermarket chain in Sri Lanka. This is where customers can order groceries online. Staff will manage orders, stocks and deliveries using the system

 Project Discussion Bot - Year2

Thanks for the details! 😊 Your project **QuickMart Ordering System** sounds very useful for a local supermarket chain in Sri Lanka. 🛒💻

This system will help: 👥 Customers to order groceries online

- 👤 Staff to manage orders and stock
- 🚚 Delivery teams to handle deliveries
- 👤 Admins to control the whole system

Now, I've created 5 client personas from QuickMart supermarket for you to talk to. Each one has a different role and view about the system:

1. **Mr. Nuwan Perera** – Admin Officer
2. **Ms. Shanika Fernando** – Inventory Manager
3. **Mr. Roshan Silva** – Regular Customer
4. **Ms. Tharushi Jayawardena** – Payment Merchant Representative
5. **Mr. Kasun Dissanayake** – Delivery Coordinator

whos in in charge of the marketing?

 Project Discussion Bot - Year2 ⓘ

👋 Hi again! At QuickMart, the person in charge of marketing is **Ms. Menaka Wijesinghe – Marketing Manager**.

Would you like to ask her some questions? 😊

     

Mr. Nuwan Perera as Admin Officer

How do you manage staff accounts and roles?
Should the system send alerts for low stock or pending orders?
What kind of user access levels do you need?



Project Discussion Bot - Year2

👋 Hello! I'm Mr. Nuwan Perera, Admin Officer at QuickMart.

Right now, I manage staff accounts using Excel sheets. It's slow and not very safe. I want the system to let me create staff accounts and give them roles easily.

Yes, the system should send alerts for low stock and pending orders. That will help us act quickly.

I need different access levels. For example, staff can only see orders, but admin can change prices and manage users. That way, everything stays secure.

Ms. Shanika Silva as Inventory Manager

How do you check if an item is out of stock?
Do you want to get alerts when stock is low?
Can customers see which items are available in real-time?



Project Discussion Bot - Year2

👋 Hello! I'm Ms. Shanika Silva, the Inventory Manager at QuickMart.

Right now, I check stock manually or ask staff to count items. It takes time and sometimes we miss low stock items.

Yes, I really want alerts when stock is low. A message or notification would help me act fast.

Customers should be able to see available items in real-time. If something is out of stock, they should not be able to order it.

Let me know if you need more help 😊

Mr. Roshan Silva,

What features do you want in the online ordering system?
How do you prefer to pay for your orders?
Do you want to track your delivery?



Project Discussion Bot - Year2

👋 Hello! I'm Mr. Roshan Silva, a regular customer at QuickMart.

I want the system to be simple and fast. I like to see product pictures, prices, and offers. I also want to save my favourite items and order them again easily.

I usually pay using my bank card or mobile wallet. It should be safe.

Yes, I want to track my delivery. It helps me know when my order will arrive.

So, Ms. Menaka Wijesinghe as the Marketing Manager?

How do you promote new products or offers to customers?

Should customers get emails or SMS about promotions?

How do you want to collect customer feedback?



Project Discussion Bot - Year2 ⓘ

👋 Hello! I'm Ms. Menaka Wijesinghe, the Marketing Manager at QuickMart.

Thanks for asking! Here's how I handle marketing:

👉 We promote new products and offers using posters in stores, social media, and sometimes flyers. But we want to do more online.

👉 Yes, I think customers should get emails or SMS about promotions. It helps them know what's new and what's on discount.

👉 For feedback, I want a simple form in the system. Customers can rate their experience and write comments. That will help us improve.

Ms. Tharushi Jayawardena as Payment Merchant Representative

Do you want the system to support online payments?

What payment methods should be available? (e.g., card, bank, mobile)

Should the system send alerts for failed payments?



Project Discussion Bot - Year2

👋 Hello! I'm Ms. Tharushi Jayawardena, Payment Merchant Representative at QuickMart.

Yes, I want the system to support online payments. It will make things faster and easier for customers.

The system should allow card payments, bank transfers, and mobile wallets. These are popular in Sri Lanka.

Also, yes — the system should send alerts if a payment fails. That way, we can help the customer quickly and avoid delays.

Mr. Kasun Dissanayake as Delivery Coordinator

How do you receive delivery orders now?

What details do you need for each delivery?

Can you update delivery status in the system?



Project Discussion Bot - Year2

👋 Hello! I'm Mr. Kasun Dissanayake, the Delivery Coordinator at QuickMart.

Right now, I get delivery orders through phone calls or printed lists. Sometimes it's confusing and slow.

For each delivery, I need the customer's name, address, phone number, order items, and delivery time. If anything is missing, it causes problems.

Yes, I would like to update the delivery status in the system. It will help everyone know if the order is on the way or delivered. That will make my work easier and faster.

Identified Users and their Roles

User	Description
Admin Officer	- Manage user accounts - Check reports and monitor user activity.
Inventory Manager	- Update item quantities - Check low stock alerts - Update the prices of the products.
Regular Customer	- Browse products - Place orders - Make payments and provide feedback
Marketing Manager	- Manage promotions to show special offers - Discounts and seasonal deals to customers
Payment Merchant Representative	- Ensure that payment services are secure and check if payments are successful or failed.
Delivery Coordinator	- Assign delivery staff - Update delivery status - Monitor delivery - Progress and respond to delays and issues.

Core Functions of the System

1. User Management
2. Product & Inventory Management
3. Order Processing System
4. Payment Management
5. Delivery Management
6. Marketing and Promotion Management

1. User management

Purpose:

- To control access and assign roles to different staff members

How it Works:

- Admins create user accounts and assign predefined roles (e.g., customer, admin, delivery staff), each with role-specific permissions.
- The system logs user activities and allows the admin to block, suspend, or reactivate users as needed.
- Passwords can be reset securely, and optional two-factor authentication can be enabled for account protection.

2. Product & Inventory Management

Purpose:

- To manage product information and maintain accurate inventory levels.

How it Works:

- The system allows inventory managers to add, update, or remove product details such as name, price, category, and stock quantity.
- Inventory levels are monitored in real time, triggering alerts when items are low or nearing expiry.
- Products can be imported or exported via Excel/CSV for bulk updates.

3. Order Processing System

Purpose:

- To allow customers to browse, shop, and place orders online easily.

How it Works:

- Customers can search, filter, and view products, then add desired items to their shopping cart or wish list.
- During checkout, users choose delivery options, apply promo codes, and confirm the order.
- Orders can be tracked by post purchase, and customers may view, reorder, or cancel orders from their order history.

4. Payment Management

Purpose:

- To enable secure, and verifiable payment transactions.

How it Works:

- The system integrates with third party payment gateways to support multiple payment modes (cards, wallets, etc.).
- Once payment is processed, receipts are generated, and customers are notified via email/SMS.
- Failed payments trigger alerts and offer retry options; refunds are processed through the same channel if needed.

5. Delivery Management

Purpose:

- To ensure accurate, timely, and efficient delivery of customer orders.

How it Works:

- Delivery managers assign orders to available delivery personnel and set delivery slots.
- Orders are tracked at each stage (packed, in transit, delivered), with updates sent to customers.
- Delivery confirmations and issues (e.g., delays or failed delivery) are managed and logged into the system.

6. Marketing and Promotion Management

Purpose:

- To attract and retain customers through targeted promotions and campaigns.

How it Works:

- Marketing staff can create and manage discount codes, bundle offers, and seasonal campaigns.
- The system sends personalized notifications and emails based on customer behavior and purchase history.
- Campaign performance is tracked via metrics such as redemption rates, click-throughs, and conversions.